



Chapter 5 Discrete Fourier Transform

—— DFT



Outline



- **Orthogonal Transforms**
 - ✓ **The Definition of DFT**
 - ✓ **The Relationship Between DFT and DTFT**
 - ✓ **The Computation of Finite-Length Sequences**
- **Classifications of Finite-Length Sequences**
 - ✓ **DFT Properties and Theorems**
- **Fourier-Domain Filtering**
- **Computation of the DFT of Real sequences**
 - ✓ **Linear Convolution Using DFT**

5.1 Orthogonal Transforms



- General form of the orthogonal transform pair:

$$X[k] = \sum_{n=0}^{N-1} x[n] \psi^*[k, n] \quad 0 \leq k \leq N-1$$

analysis equation

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] \psi[k, n] \quad 0 \leq n \leq N-1$$

synthesis equation

- $\psi[k, n]$ --- basis sequences, are also length-N sequences in both domains, which satisfy:

$$\frac{1}{N} \sum_{n=0}^{N-1} \psi[k, n] \psi^*[l, n] = \begin{cases} 1, l = k \\ 0, l \neq k \end{cases}$$

A periodic sequence $\tilde{x}[n] = \tilde{x}[n + rN]$ (r is integer, N is period) can be represented as a **Fourier series**.

The orthogonal functions set is: $\{\psi[k, n] = e^{j\left(\frac{2\pi}{N}\right)kn}\} \quad (k = 0, 1, 2, \dots, N-1)$

$$\tilde{x}[n] = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] e^{j\frac{2\pi}{N}kn} = \frac{1}{N} \sum_{k=0}^{N-1} \tilde{X}[k] W_N^{-kn} \quad \dots(\text{A})$$

$$\tilde{X}[k] = \sum_{n=0}^{N-1} \tilde{x}[n] e^{-j\frac{2\pi}{N}kn} = \sum_{n=0}^{N-1} \tilde{x}[n] W_N^{kn} \quad \dots(\text{B})$$

$$W_N = e^{-j\frac{2\pi}{N}}$$

(kernel)



Note:

① $\psi[k, n]$ has period N , $\psi[0, n] = 1$ is DC component, $\psi[k, n]$ is the k th-harmonic.

$$\textcircled{2} \quad \sum_{n=0}^{N-1} \psi[k, n] \psi^*[\ell, n] = \sum_{n=0}^{N-1} e^{j \frac{2\pi}{N} (k-\ell)n} = \frac{1 - e^{j \frac{2\pi}{N} (k-\ell)N}}{1 - e^{j \frac{2\pi}{N} (k-\ell)}} = \begin{cases} 0, & 1 \neq k \\ N, & 1 = k + rN \end{cases}$$

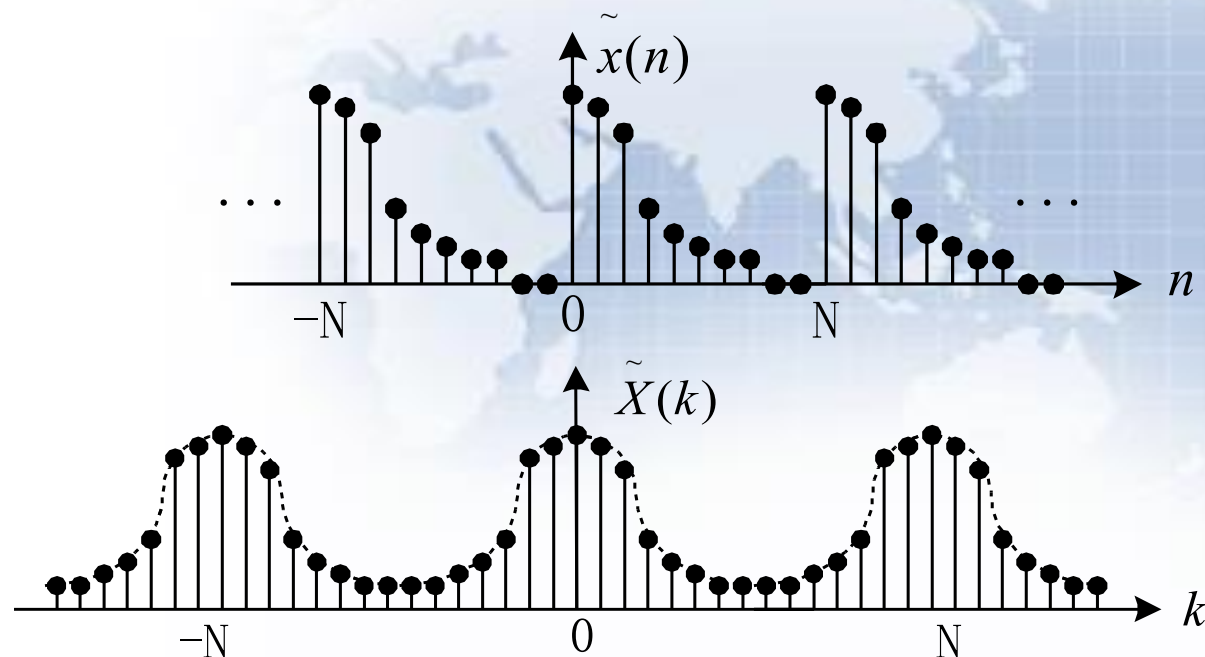
$$\textcircled{3} \quad \psi[k + N, n] = \psi[k, n] \quad \left\{ \psi[k, n] = e^{j \left(\frac{2\pi}{N} \right) kn} \right\}$$

④ The equation (A) and (B) are expressed as



$$\tilde{x}[n] \Leftrightarrow \tilde{X}[k]$$

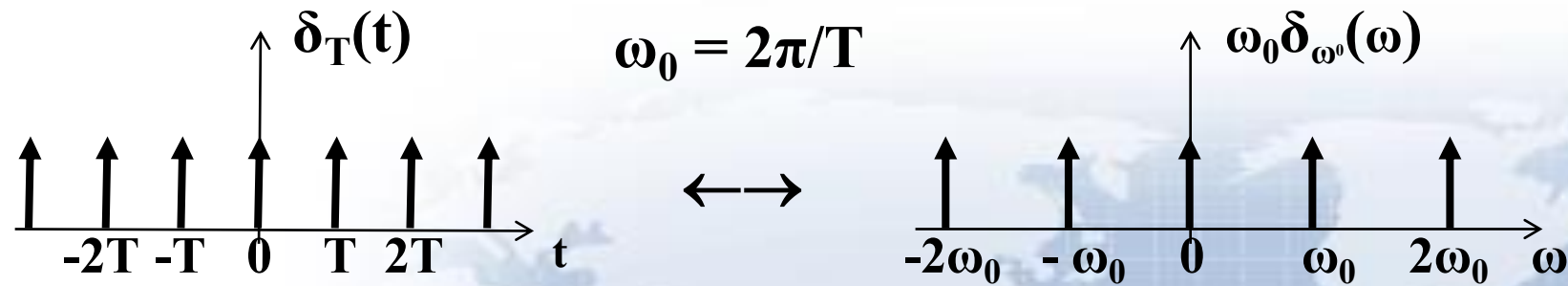
or $\tilde{X}[k] = DFS\{\tilde{x}[n]\} \quad \tilde{x}[n] = IDFS\{\tilde{X}[k]\}$



Typical DFS Pair



- $\delta_T(t) \longleftrightarrow \omega_0 \delta_{\omega_0}(\omega)$



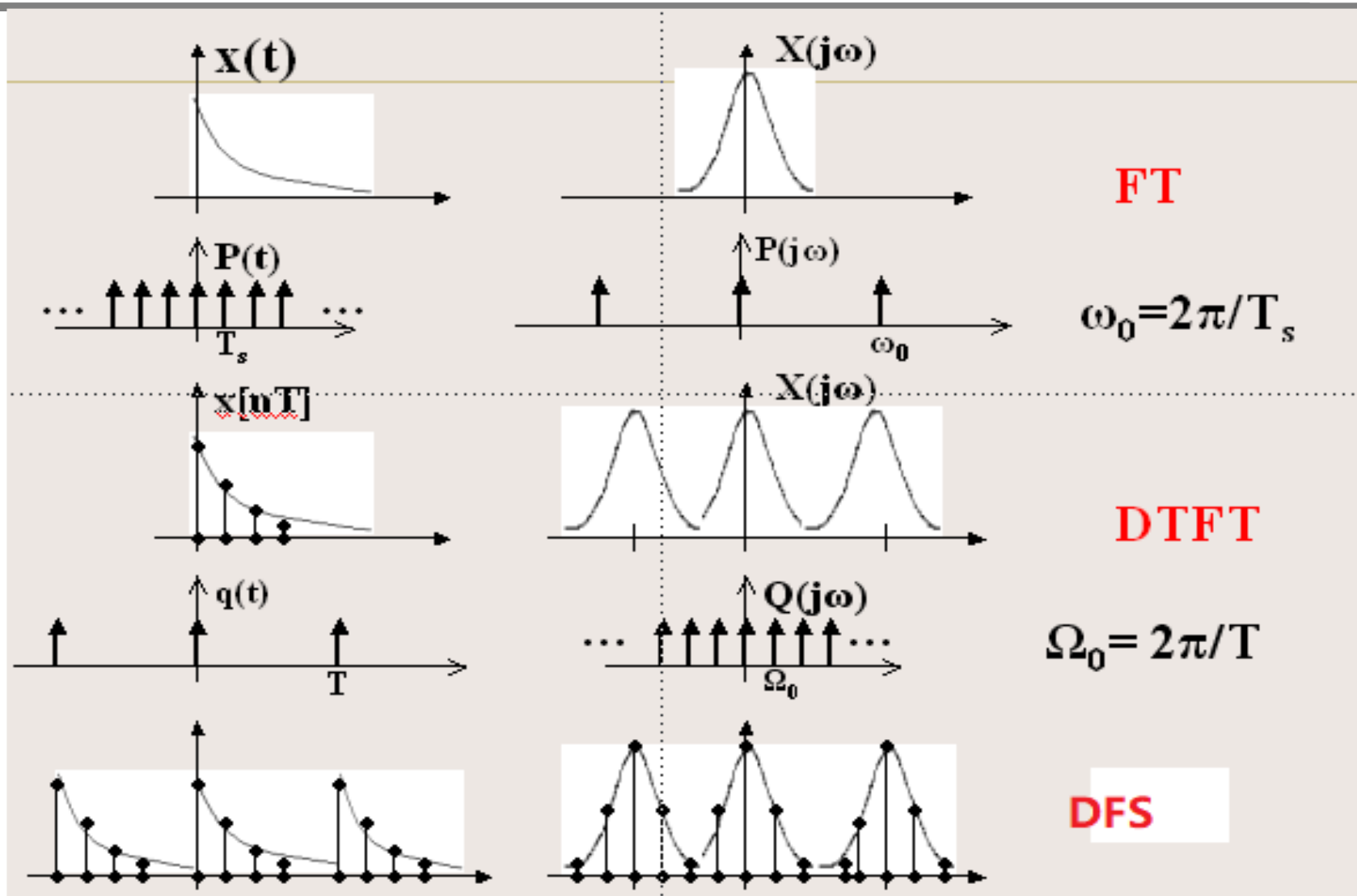
- The signals in both sides are **periodical**, so the processing could be in **one period**, which is important because:
 - (1) the number of calculation is limited, which is necessary for computer;
 - (2) all of the signal information could be kept in one period, which is necessary for accurate processing.

Make a Signal Discrete and Periodical



- The engineering signals are often continuous and aperiodical. If we want to process the signals with DFS, we have to make the signals discrete and periodical.
- **Sampling** to make the signal be discrete.
- Make the signal periodical by periodical expanding.
 - If $x[n]$ is a **limited length** N -point sequence, see it as one period of a periodical signal;
 - If $x[n]$ is an infinite length sequence, **cut-off** its tail to make a N -point sequence, then do the periodic extending. The **truncation** will introduce distortion. ---- **windowing**

Make a signal discrete and periodical



5.2 The Definition of DFT



$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, 0 \leq k \leq N-1$$

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, 0 \leq n \leq N-1$$

Where: $W_N = e^{-j2\pi/N}$

$$x[n] \underset{N}{\overset{DFT}{\longleftrightarrow}} X[k]$$

MOOC

Verify:



$$\begin{aligned} X[k] &= \sum_{n=0}^{N-1} x[n] W_N^{kn} = \sum_{n=0}^{N-1} \left(\frac{1}{N} \sum_{l=0}^{N-1} X[l] W_N^{-ln} \right) W_N^{kn} \\ &= \frac{1}{N} \sum_{n=0}^{N-1} \sum_{l=0}^{N-1} X[l] W_N^{(k-l)n} = \sum_{l=0}^{N-1} X[l] \left\{ \frac{1}{N} \sum_{n=0}^{N-1} W_N^{(k-l)n} \right\} \end{aligned}$$

$$\text{Q } \sum_{n=0}^{N-1} W_N^{(k-l)n} = \begin{cases} N, & \text{for } k-l = mN, \text{ } m \text{ an integer} \\ 0, & \text{otherwise} \end{cases}$$

Hence,
$$X[k] = \sum_{l=0}^{N-1} X[l] \delta[k-l]$$

MOOC

Example: Consider the length- N sequence



$$x[n] = \begin{cases} 1, & n = 0 \\ 0, & 1 \leq n \leq N-1 \end{cases}$$

Its N -point DFT is given by:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn} = x[0] W_N^0 = 1$$
$$0 \leq k \leq N-1$$

MOOC

Example: Consider the length- N sequence



$$g[n] = \cos(2\pi rn/N), \quad 0 \leq r \leq N-1$$

• Using a trigonometric identity we can write:

$$g[n] = \frac{1}{2} \left(e^{j2\pi rn/N} + e^{-j2\pi rn/N} \right) = \frac{1}{2} \left(W_N^{-rn} + W_N^{rn} \right)$$

$$G[k] = \sum_{n=0}^{N-1} g[n] W_N^{kn} = \frac{1}{2} \left(\sum_{n=0}^{N-1} W_N^{-(r-k)n} + \sum_{n=0}^{N-1} W_N^{(r+k)n} \right),$$

$$0 \leq k \leq N-1$$

MOOC



Making use of the identity:

$$\sum_{n=0}^{N-1} W_N^{-(k-l)n} = \begin{cases} N, & \text{for } k-l = mN, \quad m \text{ an integer} \\ 0, & \text{otherwise} \end{cases}$$

we get:

$$G[k] = \begin{cases} N/2, & \text{for } k = r \\ N/2, & \text{for } k = N - r \\ 0, & \text{otherwise} \end{cases} \quad 0 \leq k \leq N-1$$

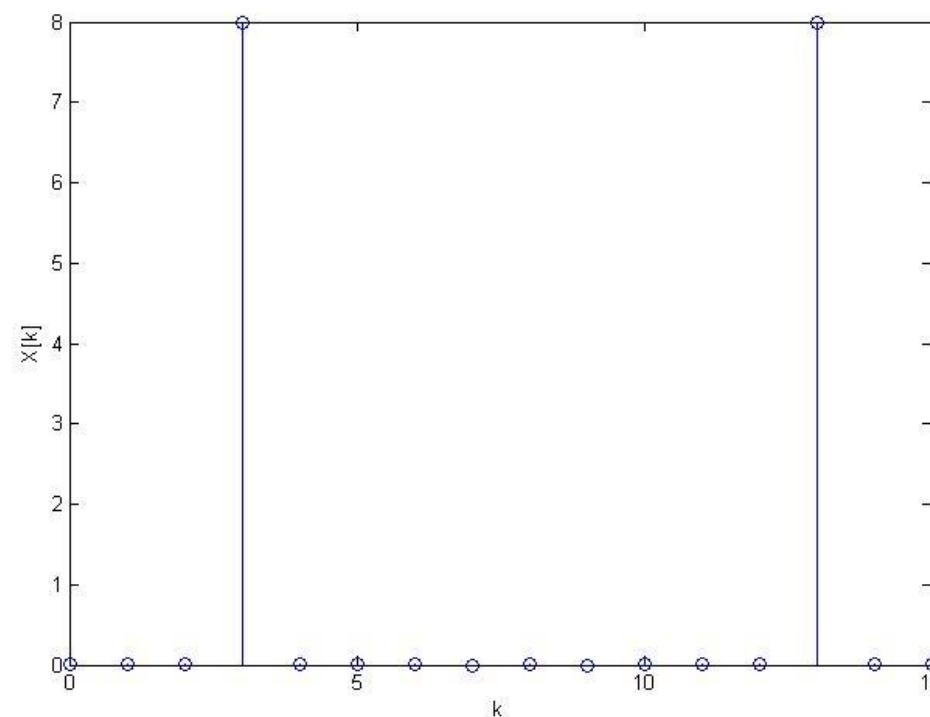
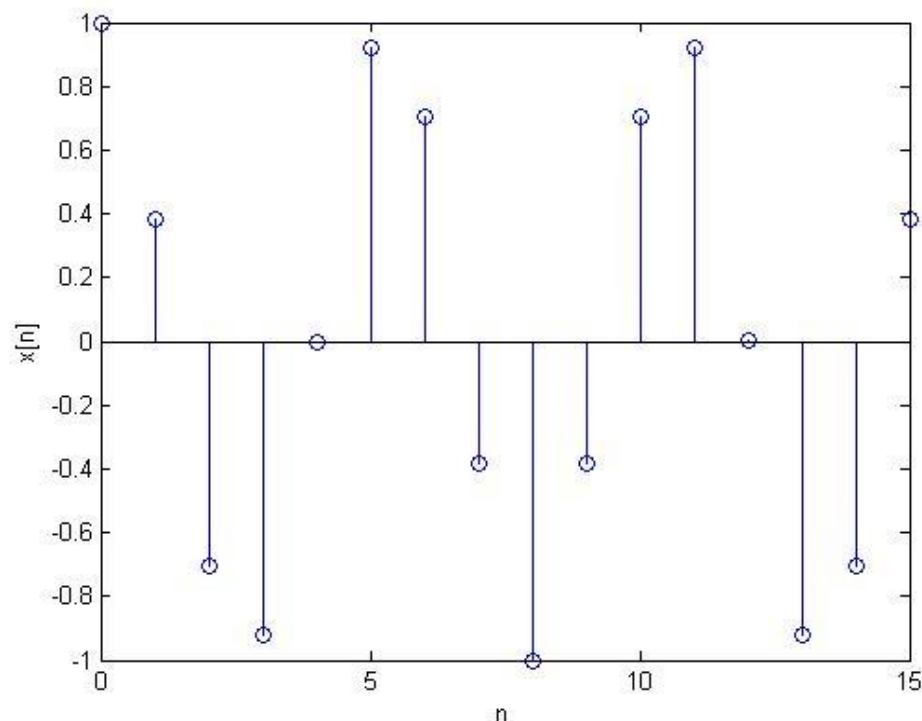
MOOC

- When $N=16$, $r=3$



$$g[n] = \cos(6\pi n/16), \quad 0 \leq n \leq 15$$

$$G[k], 0 \leq k \leq 15$$



MOOC

Matrix Relations



- The DFT samples defined by:

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{kn}, \quad 0 \leq k \leq N-1$$

Can be expressed in matrix form as $\mathbf{X} = \mathbf{D}_N \mathbf{x}$

Where,

$$\mathbf{X} = [x[0] \quad x[1] \quad \dots \quad x[N-1]]^T$$

$$\mathbf{x} = [x[0] \quad x[1] \quad \dots \quad x[N-1]]^T$$

MOOC

Matrix Relations



- And D_N is the $N \times N$ DFT matrix given by:

$$D_N = \begin{bmatrix} 1 & 1 & 1 & L & 1 \\ 1 & W_N^1 & W_N^2 & L & W_N^{(N-1)} \\ 1 & W_N^2 & W_N^4 & L & W_N^{2(N-1)} \\ M & M & M & M & M \\ 1 & W_N^{(N-1)} & W_N^{2(N-1)} & L & W_N^{(N-1)^2} \end{bmatrix}$$

MOOC

Matrix Relations



- Likewise, the IDFT relation is given by:

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn}, \quad 0 \leq k \leq N-1$$

**can be expressed in matrix form as $\mathbf{x} = \mathbf{D}_N^{-1} \mathbf{X}$,
where $\mathbf{D}_N^{-1}$ is the $N \times N$ IDFT matrix.**

MOOC

Matrix Relations



$$\mathbf{D}_N^{-1} = \frac{1}{N} \begin{bmatrix} 1 & 1 & 1 & L & 1 \\ 1 & W_N^{-1} & W_N^{-2} & L & W_N^{-(N-1)} \\ 1 & W_N^{-2} & W_N^{-4} & L & W_N^{-2(N-1)} \\ M & M & M & M & M \\ 1 & W_N^{-(N-1)} & W_N^{-2(N-1)} & L & W_N^{-(N-1)^2} \end{bmatrix}$$

Note: $\mathbf{D}_N^{-1} = (\mathbf{D}_N^*) / N$

MOOC

DFT Computation Using MATLAB

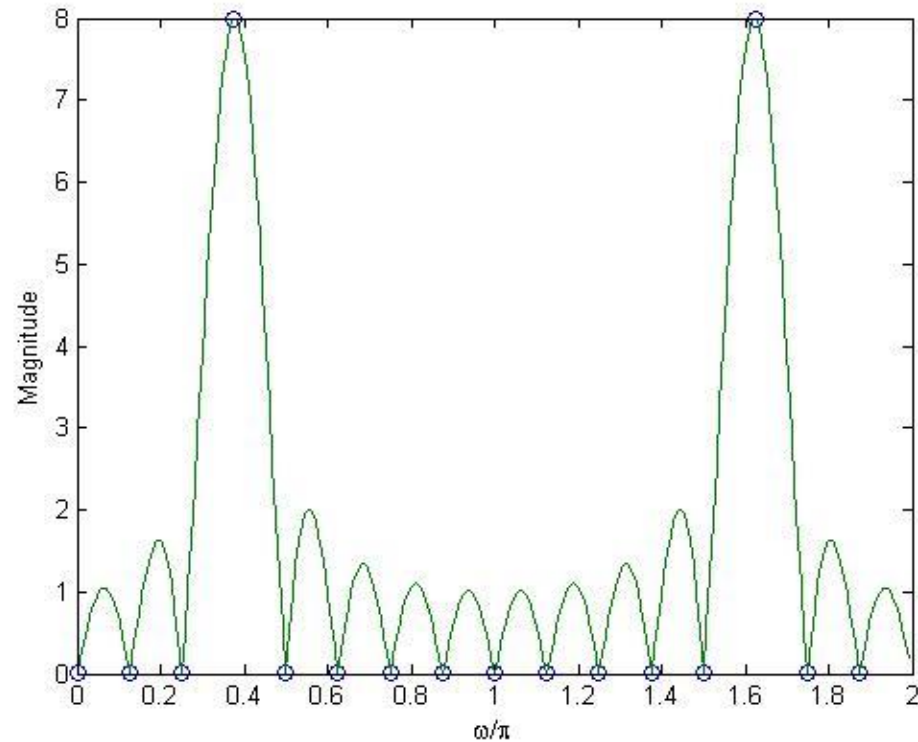


- The functions to compute the DFT and the IDFT are **FFT** and **IFFT**.
- These functions make use of FFT algorithms which are computationally highly efficient compared to the direct computation.
 - From $O(n^2)$ to $O(n\log_2 n)$
- **Programs 5_1.m** and **5_2.m** illustrate the use of these functions.



- Example - **Program 5_3.m** can be used to compute the DFT and the DTFT of the sequence:

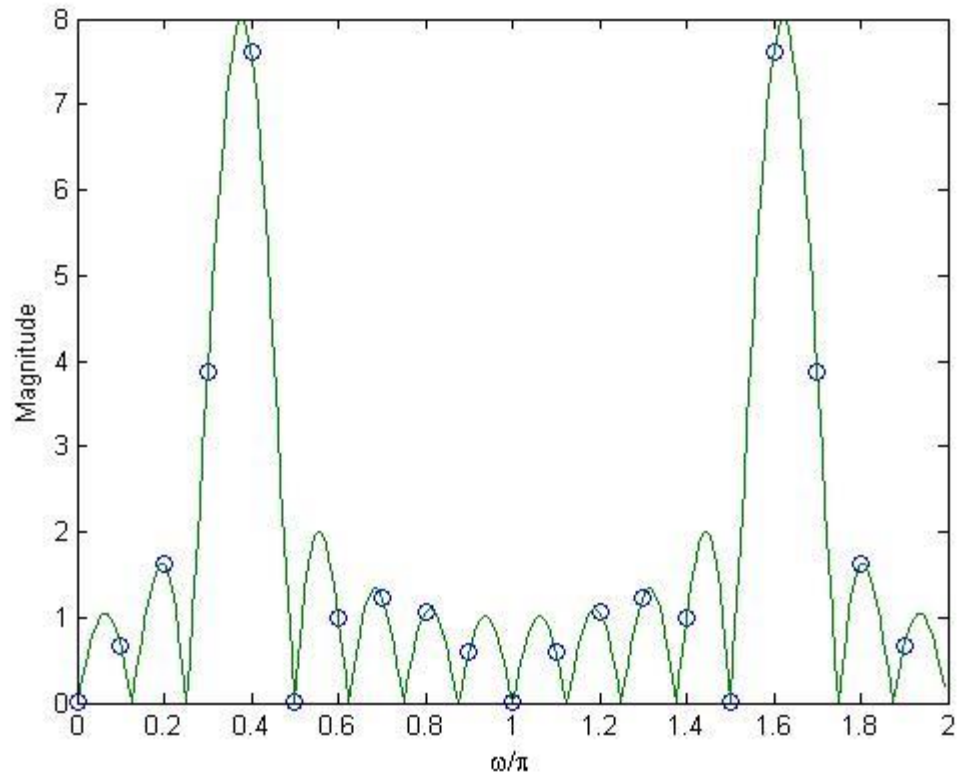
$$x[n] = \cos(6\pi n/16), \quad 0 \leq n \leq 15$$



•fft(x, 16)

○ indicates DFT samples

- $\text{fft}(x, 20)$





5.3 The Relationship Between DFT and DTFT

- *From DTFT to DFT*
- *DTFT from DFT by Interpolation*
- *Sampling the DTFT*



From DTFT to DFT

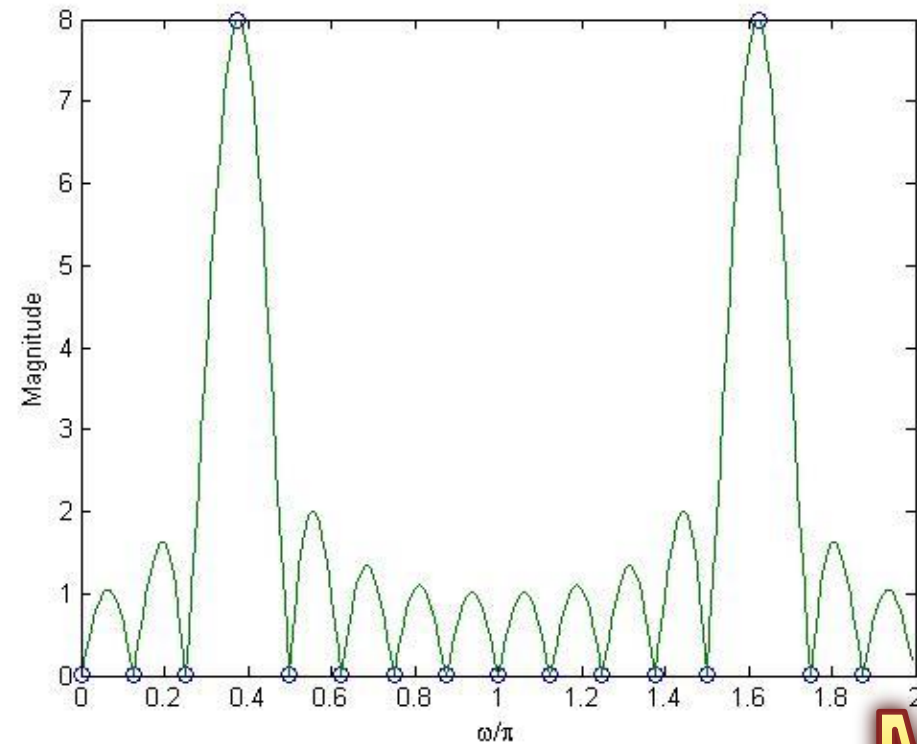


- Sampling the DTFT of a N points sequence with the space is $2\pi/N$ to get N frequency points, that is DFT.

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n} = \sum_{n=0}^{N-1} x[n]e^{-j\omega n}$$

$$X[k] = X(e^{j\omega})|_{\omega=2\pi k/N} = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}$$

$$0 \leq \omega < 2\pi, 0 \leq k \leq N-1$$

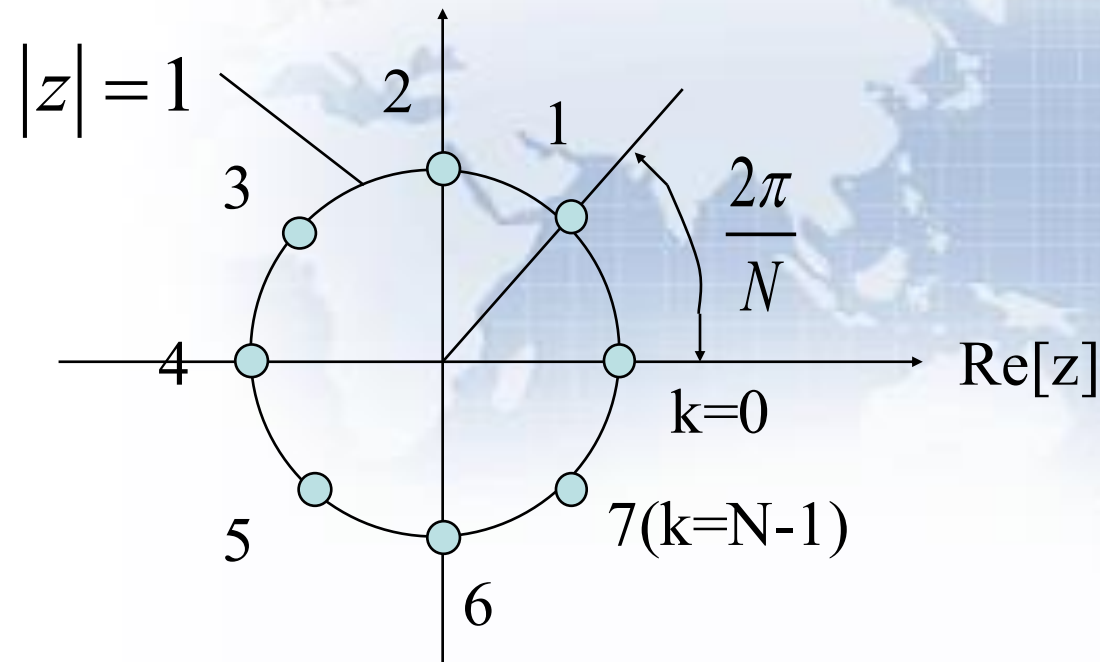


MOOC

From DTFT to DFT



- **DTFT: discrete in time domain, continuous in frequency domain.**
- **Sampling the DTFT of sequences to get N frequency points to research, that is DFT.**



Numerical Computation of the DTFT Using the DFT



- How to compute the DTFT of a length- N sequence $x[n]$ by using computer?
 1. Sampling the $X(e^{j\omega})$ to get a length- M sequence, $M \gg N$. Each frequency component is $X(e^{j2\pi/M})$.
 2. If the DFT of a sequence is length- M , the sequence must be a length- M sequence. So we build a length- M sequence use $x[n]$.
- The function **freqz** employs this approach to evaluate the frequency response at a prescribed set of frequencies of a DTFT expressed as a rational function in $e^{-j\omega}$.

$$x_e[n] = x[n], 0 \leq n \leq N-1$$

$$x_e[n] = 0, N \leq n \leq M-1$$

$$X(e^{j\omega_k}) = \sum_{n=0}^{M-1} x_e[n] e^{-j2\pi kn/M}$$

- **We can see above equation is the DFT of a length-M sequence. Because $M \gg N$, it can be seen as an approach of the DTFT of a length-N sequence.**

Sampling the DTFT



- Consider a sequence $x[n]$ with DTFT $X(e^{j\omega})$
- We sample $X(e^{j\omega})$ at N equally spaced points $\omega_k = 2\pi k/N$, $0 \leq k \leq N-1$ developing the N frequency samples $\{X(e^{j\omega_k})\}$
- These N frequency samples can be considered as an N -point DFT $Y[k]$ whose N -point IDFT is a length- N sequence $y[n]$.

What's relationship between $y[n]$ and $x[n]$?

MOOC

Sampling the DTFT



$$X(e^{j\omega}) = \sum_{\ell=-\infty}^{\infty} x[\ell]e^{-j\omega\ell}$$

Thus: $Y[k] = X(e^{j\omega_k}) = X(e^{j2\pi k/N})$

$$= \sum_{l=-\infty}^{\infty} x[l]e^{-j2\pi kl/N} = \sum_{l=-\infty}^{\infty} x[l]W_N^{kl}$$

IDFT of $Y[k]$ yields:

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} Y[k]W_N^{-kn}, 0 \leq n \leq N-1$$

MOOC

Sampling the DTFT



$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} \sum_{l=-\infty}^{\infty} x[l] W_N^{kl} W_N^{-kn} = \sum_{l=-\infty}^{\infty} x[l] \left\{ \frac{1}{N} \sum_{k=0}^{N-1} W_N^{-k(n-l)} \right\}$$

Making use of the identity:

$$\frac{1}{N} \sum_{n=0}^{N-1} W_N^{-k(n-l)} = \begin{cases} 1, & \text{for } l - n = mN \\ 0, & \text{otherwise} \end{cases} = \sum_{m=-\infty}^{\infty} \delta[n - l + mN]$$

We have: $y[n] = \sum_{m=-\infty}^{\infty} x[n + mN], \quad 0 \leq n \leq N-1$

MOOC

Sampling the DTFT



- **To apply :**
$$y[n] = \sum_{m=-\infty}^{\infty} x[n + mN], \quad 0 \leq n \leq N - 1$$

to finite-length sequences.

- If $x[n]$ is a length- M sequence with $M \leq N$, then $y[n] = x[n]$ for $0 \leq n \leq N-1$.
- If $M > N$, there is a **time-domain aliasing** of samples of $x[n]$ in generating $y[n]$, and $x[n]$ cannot be recovered from $y[n]$.

MOOC



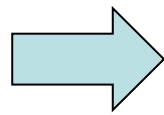
- Example: Let $\{x[n]\}=\{0 \quad 1 \quad 2 \quad 3 \quad 4 \quad 5\}$



By sampling its DTFT $X(e^{j\omega})$ at $\omega_k=2\pi k/4, 0 \leq k \leq 3$ and then applying a 4-point IDFT to these samples, we arrive at the sequence $y[n]$ given by:

$$y[n] = \sum_{m=-\infty}^{\infty} x[n+4m], \quad 0 \leq n \leq N-1$$

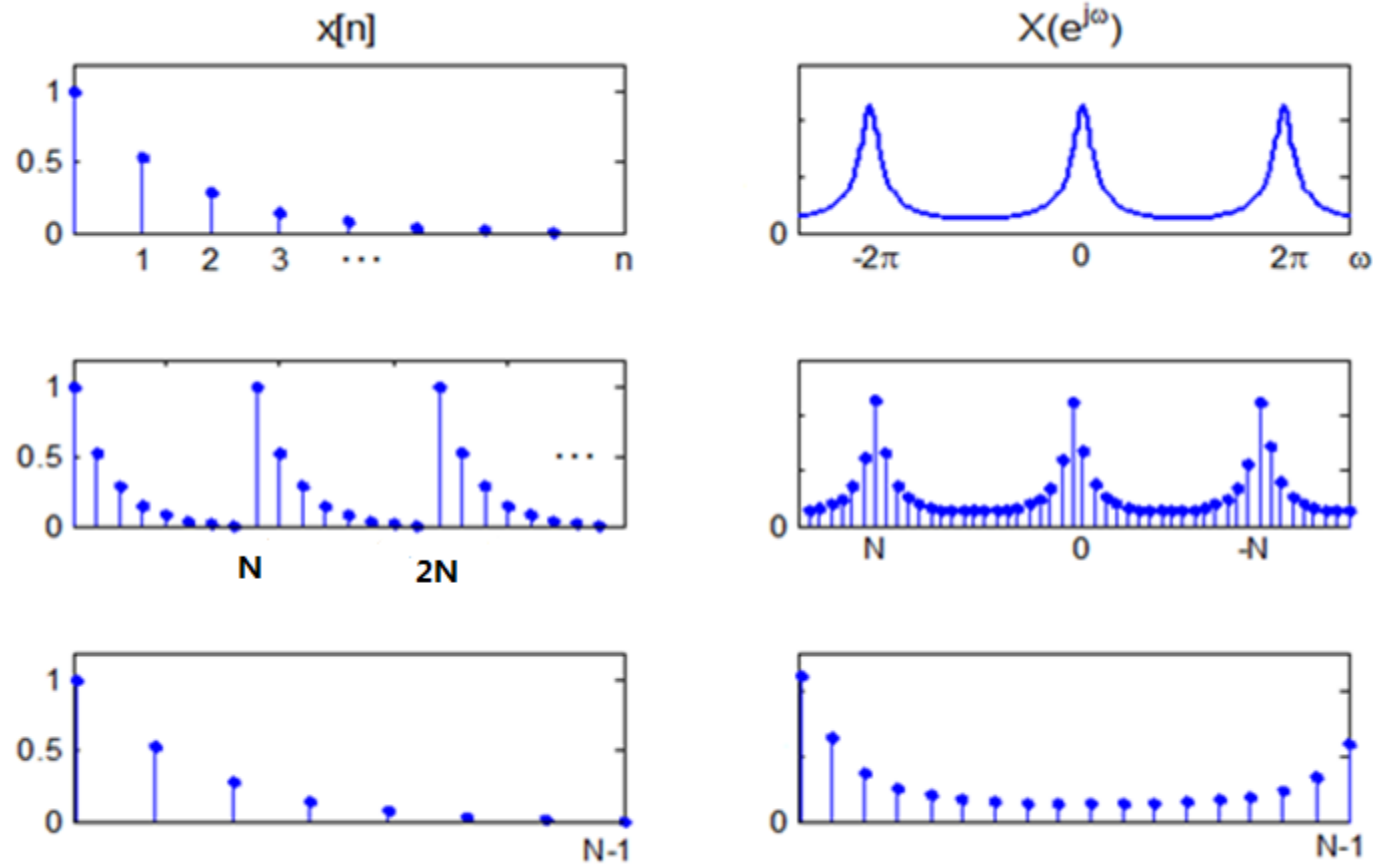
- i.e. $\{y[n]\}=\{4 \quad 6 \quad 2 \quad 3\}$



$\{x[n]\}$ cannot be recovered from $\{y[n]\}$

MOOC

Processing a signal in discrete frequency



DTFT

DFS

DFT

MOOC

Make a signal discrete and periodical



Time domain

Frequency domain

Continuous Aperiodical $\leftarrow$ **CTFT** $\rightarrow$ Continuous Aperiodical

Continuous Periodical $\leftarrow$ **FS** $\rightarrow$ Aperiodical Discrete

Discrete Aperiodical $\leftarrow$ **DTFT** $\rightarrow$ Periodical Continuous

Discrete & Periodical $\leftarrow$ **DFS** $\rightarrow$ Periodical & Discrete

MOOC

DTFT from DFT by Interpolation



- Given the N -point DFT $X[k]$ of a length- N sequence $x[n]$, its DTFT $X(e^{j\omega})$ can be uniquely determined from $X[k]$.

How?

MOOC

DTFT from DFT by Interpolation



$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=0}^{N-1} x[n] e^{-j\omega n} \\ &= \sum_{n=0}^{N-1} \left\{ \frac{1}{N} \sum_{k=0}^{N-1} X[k] W_N^{-kn} \right\} e^{-j\omega n} \\ &= \sum_{k=0}^{N-1} X[k] \left\{ \frac{1}{N} \sum_{n=0}^{N-1} e^{-j(\omega - \frac{2\pi k}{N})n} \right\} \\ &\quad \Phi_k(\omega) \end{aligned}$$

MOOC

DTFT from DFT by Interpolation



$$\Phi_k(\omega) = \frac{1}{N} \frac{1 - e^{-j(\omega - \frac{2\pi k}{N})N}}{1 - e^{-j(\omega - \frac{2\pi k}{N})}} = \frac{1}{N} \frac{\sin(\frac{\omega N - 2\pi k}{2})}{\sin(\frac{\omega N - 2\pi k}{2N})} e^{-j[(\omega - 2\pi k/N)][(N-1)/2]}$$

- **Therefore:**

$$X(e^{j\omega}) = \sum_{k=0}^{N-1} X[k] g \Phi_k(\omega)$$

- It is **interpolated** in frequency domain.

MOOC



5.4 The Computation of Finite-Length Sequences

- *Circular Shift*
- *Circular reversal*
- *Circular convolution*



Circular Shift



- **Circular Shift** of length- N sequence $x[n]$:

$$x[\langle n - n_0 \rangle_N] = \begin{cases} x[n - n_0], & \text{for } n_0 \leq n \leq N - 1 \\ x[N - n_0 + n], & \text{for } 0 \leq n \leq n_0 \end{cases}$$

$\langle \bullet \rangle_N$ — modulo operation

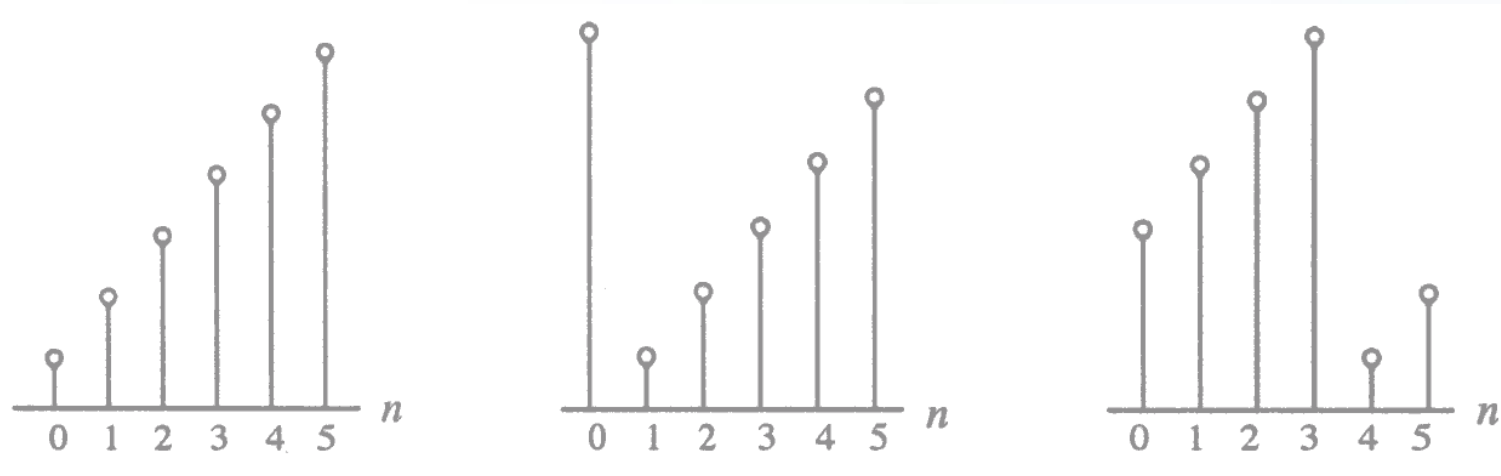
$$\langle -1 \rangle_6 = \langle 6 - 1 \rangle_6 = \langle 5 \rangle_6$$

MOOC

Circular Shift



Illustration of the concept of a circular shift:



$$x[n]$$

$$\begin{aligned} & x[\langle n - 1 \rangle_6] \\ & = x[\langle n + 5 \rangle_6] \end{aligned}$$

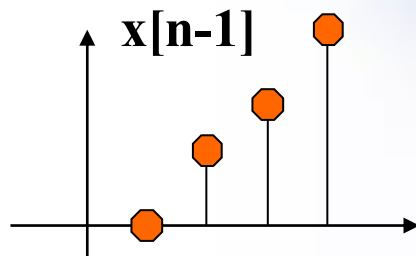
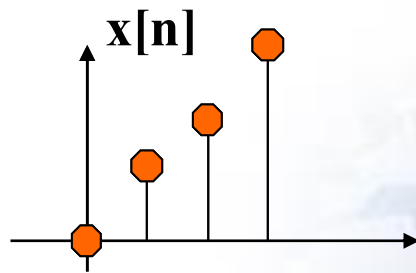
$$\begin{aligned} & x[\langle n - 4 \rangle_6] \\ & = x[\langle n + 2 \rangle_6] \end{aligned}$$

MOOC

Circular Shift of a Sequence



Non-circular shift



circular shift



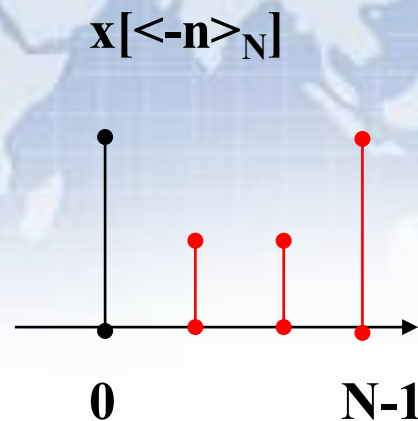
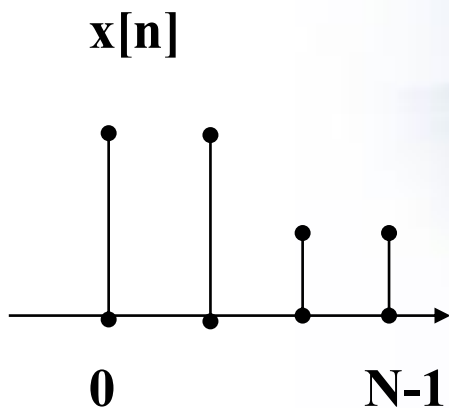
MOOC

Circular reversal



- **Circular reversal** of length- N sequence $x[n]$:

$$x[\langle -n \rangle_N] = \begin{cases} x[0], & \text{for } n = 0 \\ x[N - n], & \text{for } 1 \leq n \leq N - 1 \end{cases}$$



MOOC

Circular convolution



- To develop a convolution-like operation resulting in a length- N sequence $y_C[n]$, we need to apply circular time-reversal and circular time-shift operations.
- **Circular convolution** defined as:

$$y_C[n] = \sum_{m=0}^{N-1} g[m]h[\langle n - m \rangle_N]$$
$$= \mathbf{g[n]} \circledcirc^N \mathbf{h[n]} = \mathbf{g[n]} \circledcirc^N \mathbf{h[n]}$$

MOOC

- The N-point circular convolution can be written in matrix form as:



$$\begin{bmatrix} y_c[0] \\ y_c[1] \\ y_c[2] \\ \vdots \\ y_c[N-1] \end{bmatrix} = \begin{bmatrix} h[0] & h[N-1] & h[N-2] & \cdots & h[1] \\ h[1] & h[0] & h[N-1] & \cdots & h[2] \\ h[2] & h[1] & h[0] & \cdots & h[3] \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ h[N-1] & h[N-2] & h[N-3] & \cdots & h[0] \end{bmatrix} \begin{bmatrix} g[0] \\ g[1] \\ g[2] \\ \vdots \\ g[N-1] \end{bmatrix}$$

- The elements of each diagonal of the $N \times N$ matrix are equal
- Such a matrix is called a **circulant matrix**

MOOC

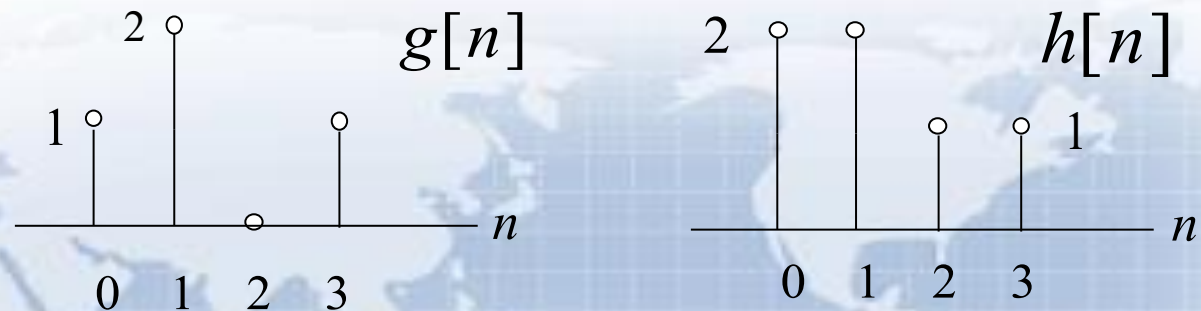
Example - Determine the 4-point circular convolution of the two length-4 sequences:



$$\{g[n]\} = \{1 \quad 2 \quad 0 \quad 1\}, \quad \{h[n]\} = \{2 \quad 2 \quad 1 \quad 1\}$$

$\uparrow \qquad \qquad \qquad \uparrow$

as sketched below:



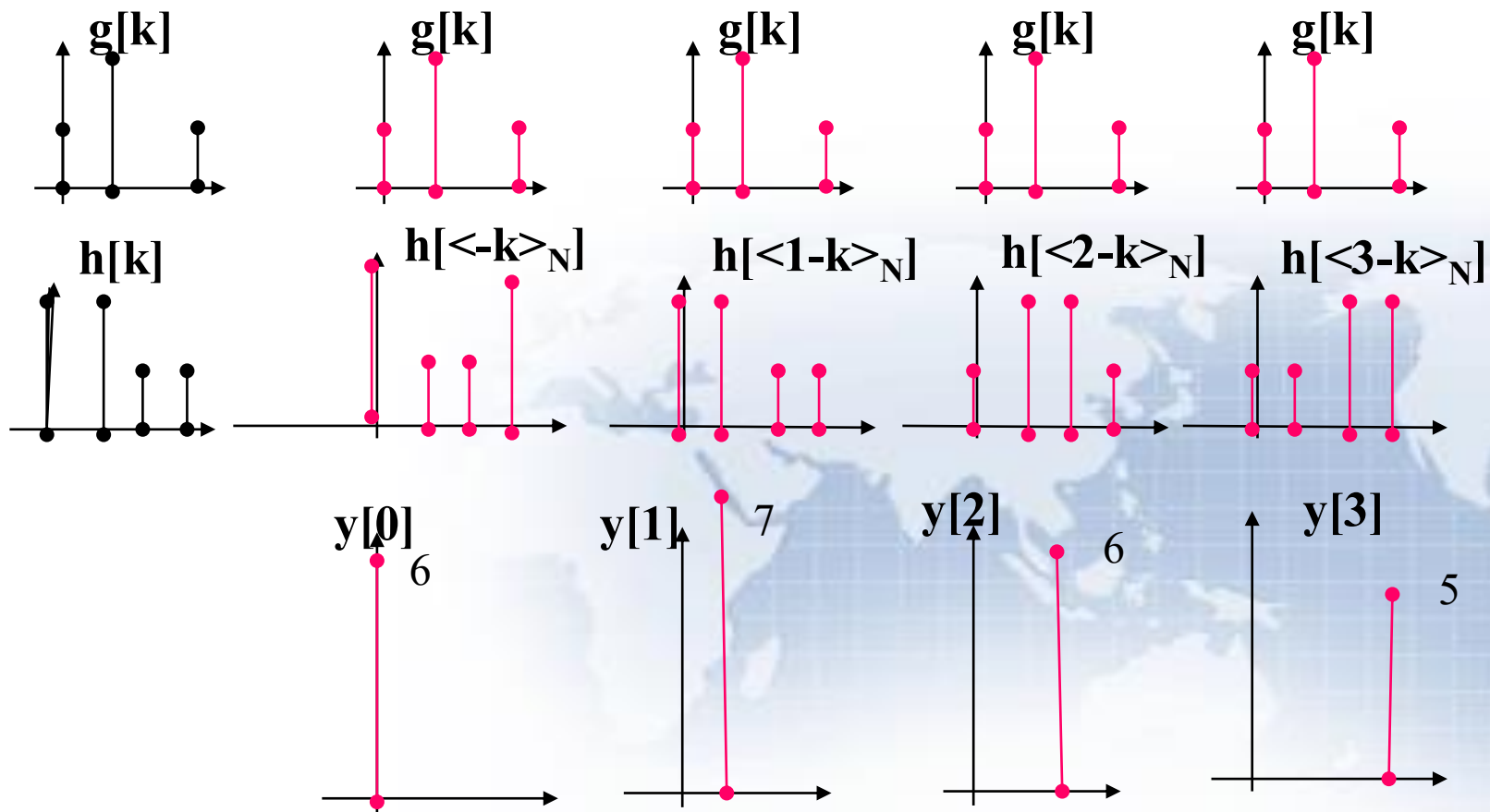
- The result is a length-4 sequence $y_C[n]$ given by:

$$y_C[n] = g[n] \textcircled{4} h[n] = \sum_{m=0}^3 g[m] h[\langle n - m \rangle_4], \quad 0 \leq n \leq 3$$

MOOC

$$\{g[n]\} = \{1 \quad 2 \quad 0 \quad 1\}, \quad \{h[n]\} = \{2 \quad 2 \quad 1 \quad 1\}$$

$\uparrow$
 $\uparrow$



$$y_c[n] = 6\delta[n] + 7\delta[n-1] + 6\delta[n-2] + 5\delta[n-3]$$

MOOC

From the above we observe:



$$\begin{aligned}y_C[0] &= \sum_{m=0}^3 g[m] h[\langle -m \rangle_4] \\&= g[0]h[0] + g[1]h[3] + g[2]h[2] + g[3]h[1] \\&= (1 \times 2) + (2 \times 1) + (0 \times 1) + (1 \times 2) = 6 \\y_C[1] &= \sum_{m=0}^3 g[m] h[\langle 1-m \rangle_4] \\&= g[0]h[1] + g[1]h[0] + g[2]h[3] + g[3]h[2] \\&= (1 \times 2) + (2 \times 2) + (0 \times 1) + (1 \times 1) = 7\end{aligned}$$



$$\begin{aligned}y_C[2] &= \sum_{m=0}^3 g[m] h[\langle 2-m \rangle_4] \\&= g[0]h[2] + g[1]h[1] + g[2]h[0] + g[3]h[3] \\&= (1 \times 1) + (2 \times 2) + (0 \times 2) + (1 \times 1) = 6\end{aligned}$$

$$\begin{aligned}y_C[3] &= \sum_{m=0}^3 g[m] h[\langle 3-m \rangle_4] \\&= g[0]h[3] + g[1]h[2] + g[2]h[1] + g[3]h[0] \\&= (1 \times 1) + (2 \times 1) + (0 \times 2) + (1 \times 2) = 5\end{aligned}$$

$$\mathbf{y[n]=6\delta[n]+7\delta[n-1]+6\delta[n-2]+5\delta[n-3]}$$

Tabular Method in Circular convolution



Example: We want to obtain the convolution sum $y_c[n]$ by the following step: $y_c[n]=g[n] \textcircled{*} h[n]$

n:	0	1	2	3	4	5
g[n]:	g[0]	g[1]	g[2]			
h[n]:	h[0]	h[1]	h[2]			
	g[0]h[0]	g[1]h[0]	g[2]h[0]			
		g[2]h[1]	g[0]h[1]	g[1]h[1]	g[2]h[1]	
		g[1]h[2]	g[2]h[2]	g[0]h[2]	g[1]h[2]	g[2]h[2]
y _c [n]:	y _c [0]	y _c [1]	y _c [2]			

MOOC

The comparison of linear convolution & circular convolution



- **The circular convolution:**

$$y_C[n] = \sum_{m=0}^{N-1} g[m]h[\langle n - m \rangle_N]$$

- **The linear convolution:**

$$y_L[n] = \sum_{m=0}^{N-1} g[m]h[n - m]$$

Please look at Fig5.6

5.5 Classifications of Finite-Length Sequences



- The **circular conjugate-symmetric** sequence and **circular conjugate-antisymmetric** sequence;
- The **geometric symmetric** sequence and **geometric antisymmetric** sequence;
- Four types of geometric symmetry sequences.

5.5.1 Classification Based on Conjugate Symmetry



- **circular conjugate-symmetry:**

$$x[n] = x^*[\langle -n \rangle_N] = x^*[\langle N - n \rangle_N], 0 \leq n \leq N - 1$$

- **circular conjugate-antisymmetry:**

$$x[n] = -x^*[\langle -n \rangle_N] = -x^*[\langle N - n \rangle_N], 0 \leq n \leq N - 1$$

MOOC



- A complex length- N sequence $x[n]$ can be expressed as:

$$x[n] = x_{cs}[n] + x_{ca}[n], 0 \leq n \leq N-1$$

where, $x_{cs}[n]$ is its **circular conjugate-symmetric part**, and $x_{ca}[n]$ is its **circular conjugate-antisymmetric part**, defined by:

$$x_{cs}[n] = \frac{1}{2} (x[n] + x^*[\langle -n \rangle_N]), 0 \leq n \leq N-1$$

$$x_{ca}[n] = \frac{1}{2} (x[n] - x^*[\langle -n \rangle_N]), 0 \leq n \leq N-1$$

MOOC



- For a real sequence $x[n]$, it can be expressed as:

$$x[n] = x_{ev}[n] + x_{od}[n], 0 \leq n \leq N-1$$

Where, $x_{ev}[n]$ is its **circular even part**, and $x_{ca}[n]$ is its **circular odd part**, defined by:

$$x_{ev}[n] = \frac{1}{2} (x[n] + x[\langle -n \rangle_N])$$

$$x_{od}[n] = \frac{1}{2} (x[n] - x[\langle -n \rangle_N])$$

MOOC

5.5.2 Classification Based on Geometric Symmetry



- Two types of geometric symmetries are usually defined:

(1) Symmetric sequence:

$$x[n] = x[N - 1 - n]$$

(2) Antisymmetric sequence:

$$x[n] = -x[N - 1 - n]$$

Since the length N of a sequence can be either even or odd, four types of geometric symmetry are defined.

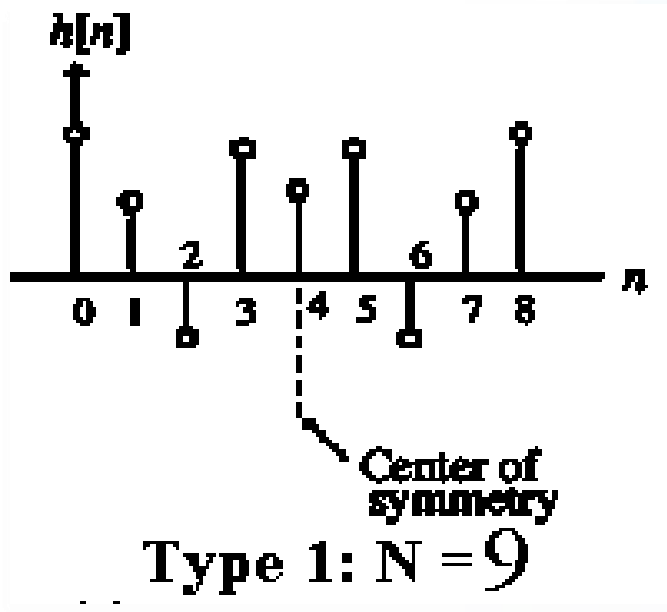
Four types of geometric symmetry sequences



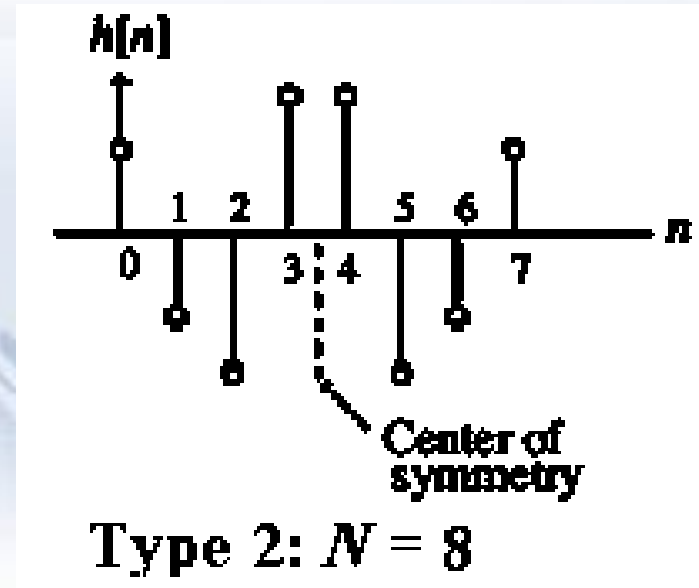
- **Type1**: Symmetric impulse response with odd length.
- **Type2**: Symmetric impulse response with even length.
- **Type3**: Antisymmetric impulse response with odd length.
- **Type4**: Antisymmetric impulse response with even length.

● In the general case for Type1 & Type2 FIR filters, the phase function is of the form:

$$\theta(\omega) = -\frac{N-1}{2}\omega + \beta$$



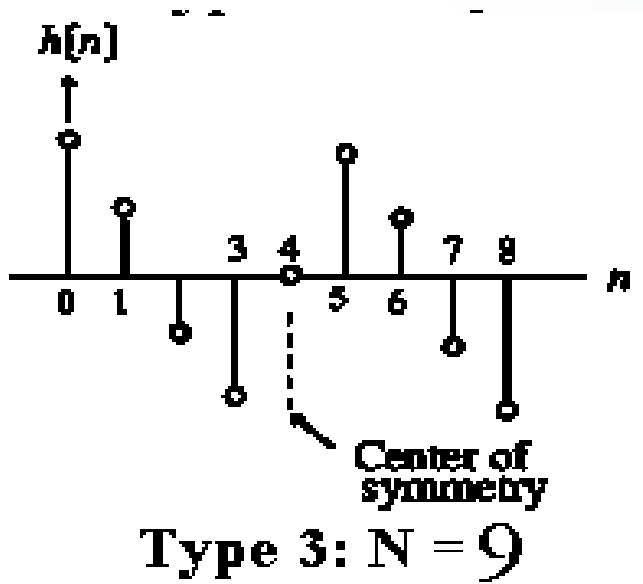
$$\theta(\omega) = -4\omega + \beta$$



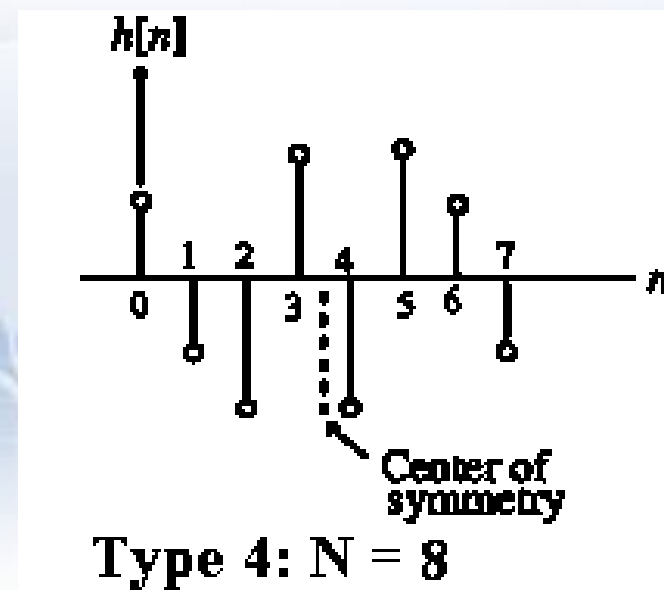
$$\theta(\omega) = -\frac{7}{2}\omega + \beta$$

● In the general case for Type3 & Type4 FIR filters, the phase function is of the form:

$$\theta(\omega) = -\frac{N-1}{2}\omega + \frac{\pi}{2} + \beta$$



$$\theta(\omega) = -4\omega + \frac{\pi}{2} + \beta$$



$$\theta(\omega) = -\frac{7}{2}\omega + \frac{\pi}{2} + \beta$$

5.6 DFT Properties

Table 5.1: Symmetric Relations 1



Length-N Sequence	N-point DFT
$x[n]$	$X[k]$
$x^*[n]$	$X^*[\langle -k \rangle_N]$
$x^*[\langle -n \rangle_N]$	$X^*[k]$
$\text{Re}\{x[n]\}$	$X_{cs}[k] = \{X[\langle k \rangle_N] + X^*[\langle -k \rangle_N]\} / 2$
$j\text{Im}\{x[n]\}$	$X_{ca}[k] = \{X[\langle k \rangle_N] - X^*[\langle -k \rangle_N]\} / 2$
$x_{cs}[n]$	$\text{Re}\{X[k]\}$
$x_{ca}[n]$	$j\text{Im}\{X[k]\}$

MOOC

Note: $x[n]$ is a **complex** sequence. $x_{cs}[n]$ and $x_{ca}[n]$ are the circular conjugate-symmetric and antisymmetric parts of $x[n]$, respectively.

Table 5.2: Symmetric Relations 2



Length-N Sequence	N-point DFT
Real $x[n]$	$X[k] = \text{Re}\{X[k]\} + j\text{Im}\{X[k]\}$
$x_{\text{ev}}[n]$	$\text{Re}\{X[k]\}$
$x_{\text{od}}[n]$	$j\text{Im}\{X[k]\}$
Symmetry relations	$X[k] = X^*[\langle -k \rangle_N]$
	$\text{Re}X[k] = \text{Re}X[\langle -k \rangle_N]$
	$\text{Im}X[k] = -\text{Im}X[\langle -k \rangle_N]$
	$ X[k] = X[\langle -k \rangle_N] $
	$\arg X[k] = -\arg X[\langle -k \rangle_N]$

Note: $x[n]$ is a **real** sequence. $x_{\text{ev}}[n]$ and $x_{\text{od}}[n]$ are the circular even and odd parts of $x[n]$, respectively.

MOOC



Example: Consider the length-14 **real sequence**

$x[n]$ has 14-point DFT $X[k]$.

If $X[2] = -1 + j3$

Then $X[12] = X^*[2] = -1 - j3$

$$X[12] = X[\langle -2 \rangle_N] = X[\langle N - 2 \rangle_N]$$

(N=14)

MOOC

5.7 DFT Theorems



Type of Property	length-N sequence	N-point DFT
	$g[n]$	$G[k]$
	$h[n]$	$H[k]$
Linearity	$ag[n]+bh[n]$	$aG[k]+bH[k]$
Circular Time-shifting	$g[\langle n-n_0 \rangle_N]$	$W_N^{kn_0}G[k]$
Frequency-shifting	$W_N^{-k_0n}g[n]$	$G[\langle k-k_0 \rangle_N]$
Duality	$G[n]$	$N[g\langle -k \rangle_N]$
Circular Convolution	$\sum_{m=0}^{N-1} g[m]h[\langle n-m \rangle_N]$	$G[k]H[k]$
Modulation	$g[n]h[n]$	$\frac{1}{N} \sum_{m=0}^{N-1} G[m]H[\langle k-m \rangle_N]$
Parseval's relation	$\sum_{n=0}^{N-1} x[n] ^2 = \frac{1}{N} \sum_{k=0}^{N-1} X[k] ^2$	

MOOC

Example: Consider the length- N sequence



$$y[n] = \begin{cases} 1, & n = m \\ 0, & 0 \leq n \leq m-1, m+1 \leq n \leq N-1 \end{cases}$$

$$\text{Q } x[n] = \begin{cases} 1, & n = 0 \\ 0, & 1 \leq n \leq N-1 \end{cases}, X[k] = 1, 0 \leq k \leq N-1$$

$$\text{and } y[n] = x[\langle n - m \rangle_N]$$

$$\therefore Y[k] = X[k]W_N^{km} = W_N^{km}, 0 \leq k \leq N-1$$

MOOC



Example - Consider the two length-4 sequences:

$$\{g[n]\} = \{1 \quad 2 \quad 0 \quad 1\}, \quad \{h[n]\} = \{2 \quad 2 \quad 1 \quad 1\}$$

$\uparrow \qquad \qquad \qquad \uparrow$

$$y_c[n] = g[n] \oplus_4 h[n], \quad 0 \leq n \leq 3$$

- **We can use DFT to compute the $y_c[n]$.**
- **The two 4-point DFTs of $g[n]$ and $h[n]$ can be computed using the matrix relation.**

MOOC

$$\begin{bmatrix} G[0] \\ G[1] \\ G[2] \\ G[3] \end{bmatrix} = D_4 \begin{bmatrix} g[0] \\ g[1] \\ g[2] \\ g[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 1-j \\ -2 \\ 1+j \end{bmatrix}$$

$$\begin{bmatrix} H[0] \\ H[1] \\ H[2] \\ H[3] \end{bmatrix} = D_4 \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 2 \\ 2 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 6 \\ 1-j \\ 0 \\ 1+j \end{bmatrix}$$

D_4 is the 4-point DFT matrix.

MOOC

- If $Y_C[k]$ denotes the 4-point DFT of $y_C[n]$, then:

$$Y_C[k] = G[k]H[k], 0 \leq k \leq 3$$

- Thus:

$$\begin{bmatrix} Y_C[0] \\ Y_C[1] \\ Y_C[2] \\ Y_C[3] \end{bmatrix} = \begin{bmatrix} G[0]H[0] \\ G[1]H[1] \\ G[2]H[2] \\ G[3]H[3] \end{bmatrix} = \begin{bmatrix} 24 \\ -j2 \\ 0 \\ j2 \end{bmatrix}$$

MOOC

- So $y_c[n]$ is 4-point IDFT of $Y_c[k]$:

$$\begin{bmatrix} y_c[0] \\ y_c[1] \\ y_c[2] \\ y_c[3] \end{bmatrix} = \frac{1}{4} \mathbf{D}_4^* \begin{bmatrix} Y_c[0] \\ Y_c[1] \\ Y_c[2] \\ Y_c[3] \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & j & -1 & -j \\ 1 & -1 & 1 & -1 \\ 1 & -j & -1 & j \end{bmatrix} \begin{bmatrix} 24 \\ -j2 \\ 0 \\ j2 \end{bmatrix} = \begin{bmatrix} 6 \\ 7 \\ 6 \\ 5 \end{bmatrix}$$

MOOC

5.8 Fourier-Domain Filtering



- Often one is interested in removing the components of a finite-length discrete-time signal in one or more frequency bands.
- A **straightforward approach** to remove the unwanted components from a signal is to implement the **filtering in the Fourier domain**.
- The Fourier-domain filtering is usually implemented using DFT. However, the DFT-based filtering will always lead to some small ripples in the filtered response.

Please look at P232, Fig5.10

Fourier-Domain Filtering



Based on DTFT convolution theorem:

$$\sum_{m=0}^{N-1} g[m]h[n-m] \longleftrightarrow G(e^{j\omega})H(e^{j\omega})$$

- In practical condition, $g[n]$ and $h[n]$ are **real-valued signal**, so the **imaginary parts** of the samples of the IDTFT of the product of their FT will be theoretically all **zeros**.
- In practice, the imaginary parts are very small numbers due to computational errors.
- The real part of the IDTFT of the product -- $y[n]$ is kept as the filtered response.

MOOC

5.9 Computation of the DFT of Real sequences



- In most practical applications, sequences are real.
- In DFT definition, the sequence is assumed to be complex.
- The DFTs of two N -point real sequences can be computed from a length- N complex sequences.
- And a $2N$ -point DFT of a length- $2N$ real sequence can be determined from a length- N complex sequences.



5.9.1 N-point DFTs of two real sequence using a single N-point DFT

- Let $g[n]$ and $h[n]$ be two real sequences of length N each, with $G[K]$ and $H[k]$ denoting their N -point DFTs.

$$x[n] = g[n] + jh[n]$$

- We build:

$$H[k] = \frac{1}{2j} \{X[k] - X^*[\langle -k \rangle_N]\}$$

- Then, we have:

$$G[k] = \frac{1}{2} \{X[k] + X^*[\langle -k \rangle_N]\}$$

5.9.2 $2N$ -Point DFT of a Real Sequences Using a Single N -Point DFT



- Let $v[n]$ be a real sequence of length $2N$ with $V[k]$ denoting its $2N$ -point DFT.
- We can define two real sequences $g[n]$ and $h[n]$ of length N each as:

$$g[n] = v[2n], h[n] = v[2n + 1]$$

- So,

$$V[k] = G[\langle k \rangle_N] + W_{2N}^k H[\langle k \rangle_N]$$

5.10 Linear Convolution Using DFT



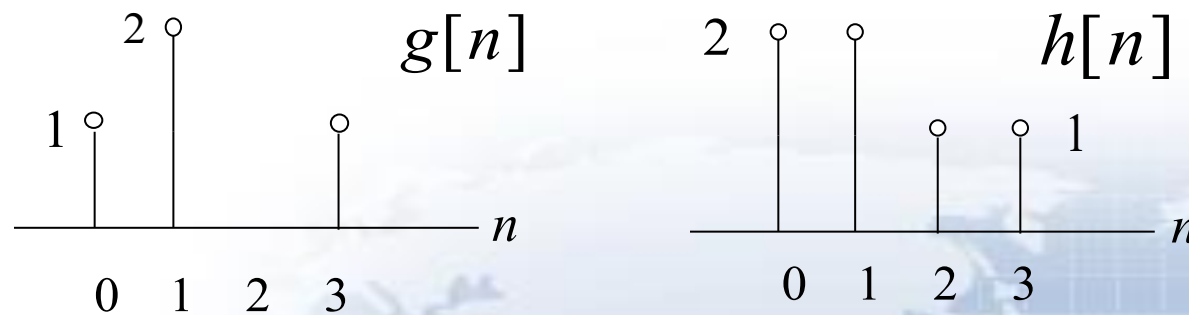
- Linear convolution is a key operation in many signal processing applications.
- Since a DFT can be efficiently implemented using FFT algorithms, it is of interest to develop methods for the implementation of linear convolution using the DFT.
- How to do?
 - Linear Convolution of Two Finite-Length Sequences;
 - Linear Convolution of a Finite-Length Sequence with an Infinite-Length Sequence.

MOOC

5.10.1 Linear Convolution of Two Finite-Length Sequences



Example: Given two length-4 sequences:



Let's extended the two length-4 sequences to length 7 by appending each with three zero-valued samples, i.e.

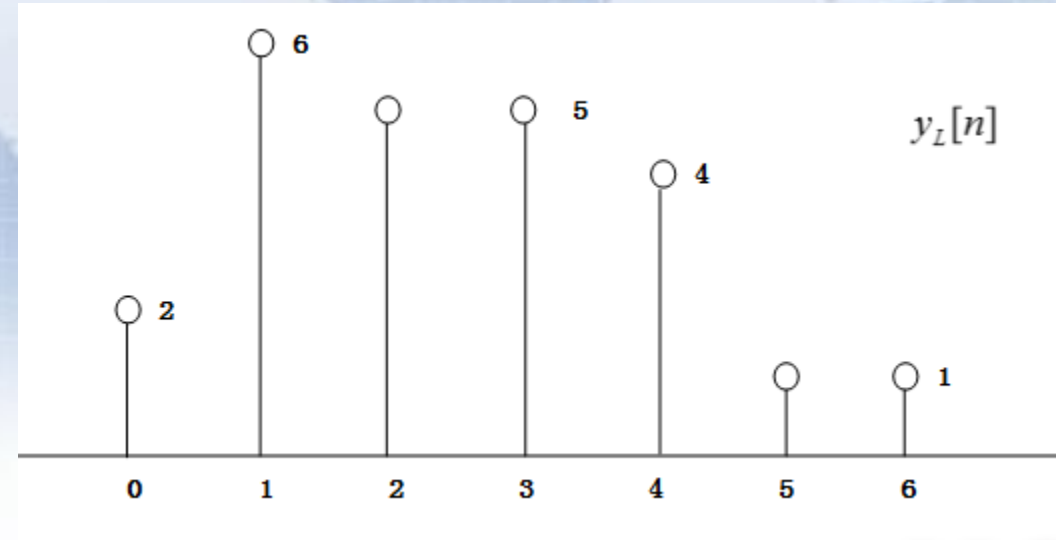
$$g_e[n] = \begin{cases} g[n], & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 6 \end{cases} \quad h_e[n] = \begin{cases} h[n], & 0 \leq n \leq 3 \\ 0, & 4 \leq n \leq 6 \end{cases}$$

MOOC

- Now determine the 7-point circular convolution of $g_e[n]$ and $h_e[n]$:

$$y_c[n] = \sum_{m=0}^6 g_e[m] h_e[\langle n - m \rangle_7], \quad 0 \leq n \leq 6$$

- As can be seen: $y_c[n]$ is same as the sequence $y_L[n]$, which is the linear convolution of $g[n]$ and $h[n]$.



- Let $g[n]$ and $h[n]$ be two finite-length sequences of length N and M , respectively.
- Define two length- L ($L=N+M-1$) sequences:

$$g_e[n] = \begin{cases} g[n], & 0 \leq n \leq N-1 \\ 0, & N \leq n \leq L-1 \end{cases}$$

$$h_e[n] = \begin{cases} h[n], & 0 \leq n \leq M-1 \\ 0, & M \leq n \leq L-1 \end{cases}$$

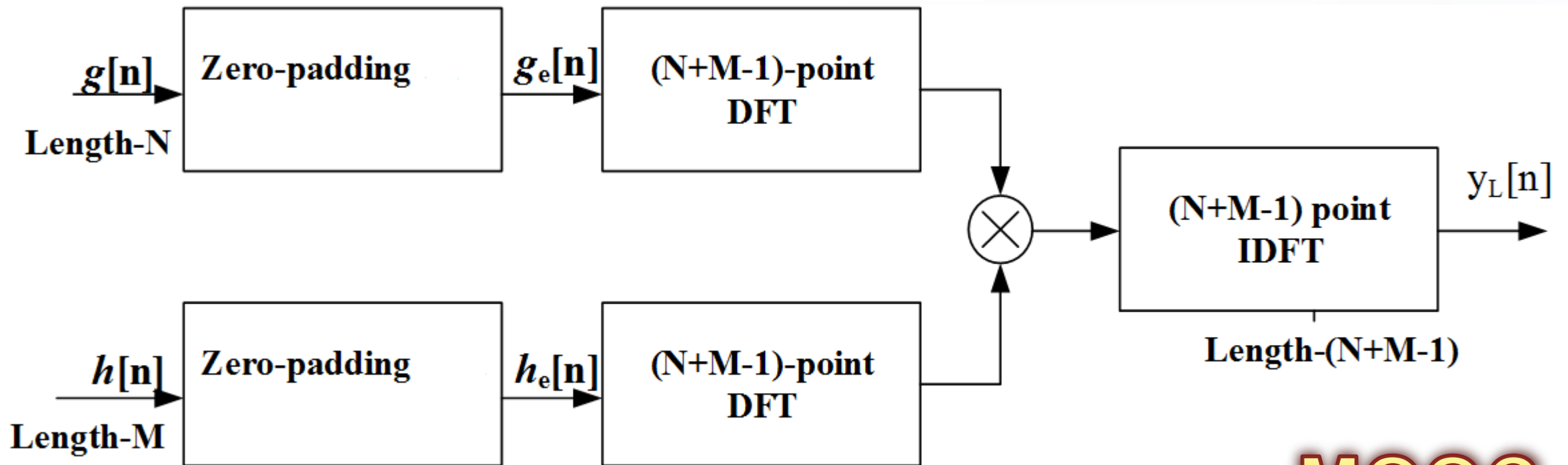
- Then $y_L[n] = g[n] * h[n] = g_e[n] \textcircled{L} h_e[n]$

MOOC

Linear Convolution Using DFT



- The corresponding implementation scheme is:



MOOC

5.10.2 The Cyclic Prefix



- In some applications, it is required to compute only a portion of the sequence $\{y[n]\}$ of length N .
- Using an N -point DFT and IDFT by appending the longer sequence with a subsequence called the **cyclic prefix**.
- The cyclic prefix plays an important role in multicarrier-based digital communication.

Example 5.18

5.10.3 Linear Convolution of a Finite-Length Sequence with an Infinite-Length Sequence



- Consider the DFT-based implementation of:

$$y[n] = \sum_{l=0}^{M-1} h[l] x[n-l] = h[n] \circledast x[n]$$

where $h[n]$ is a finite-length sequence of length M and $x[n]$ is an **infinite** length (or a finite length sequence of length much greater than M).

- There are two methods to solve this problem:
 - Overlap-Add
 - Overlap-Save

Overlap-Add Method



- Segment $x[n]$ into a set of contiguous finite-length subsequences $x_m[n]$ of length N each:

$$x[n] = \sum_{m=0}^{\infty} x_m[n - mN] \quad x_m[n] = \begin{cases} x[n + mN], & 0 \leq n \leq N-1 \\ 0, & \text{otherwise} \end{cases}$$

- Then:

$$y[n] = h[n] \circledast x[n] = \sum_{m=0}^{\infty} y_m[n - mN]$$

where $y_m[n] = h[n] \circledast x_m[n]$

Overlap-Add Method

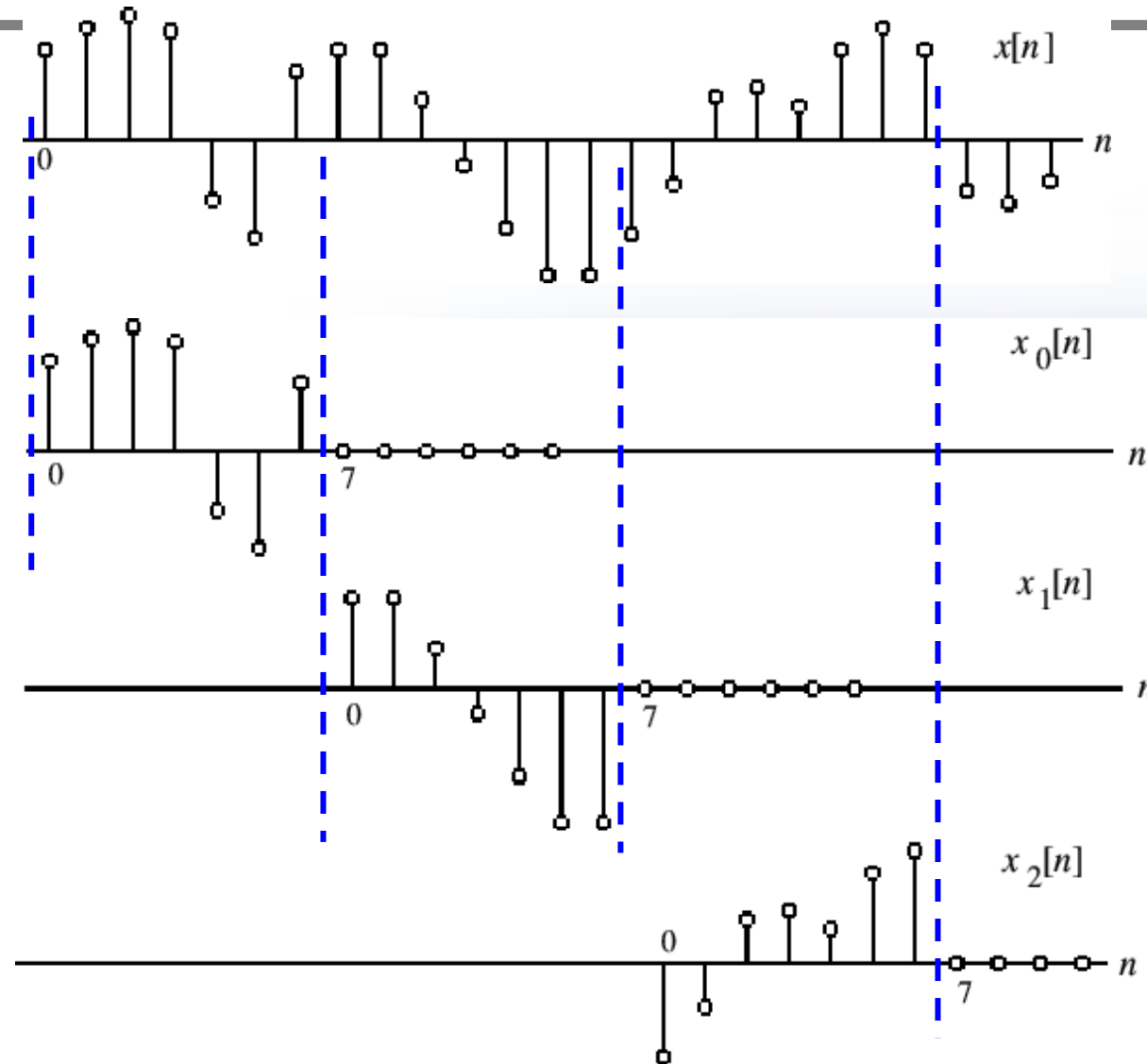


- As a result, the desired linear convolution $y[n] = h[n] \circledast x[n]$ has been broken up into a sum of infinite number of **short-length linear convolutions** of length **$N+M-1$** each:

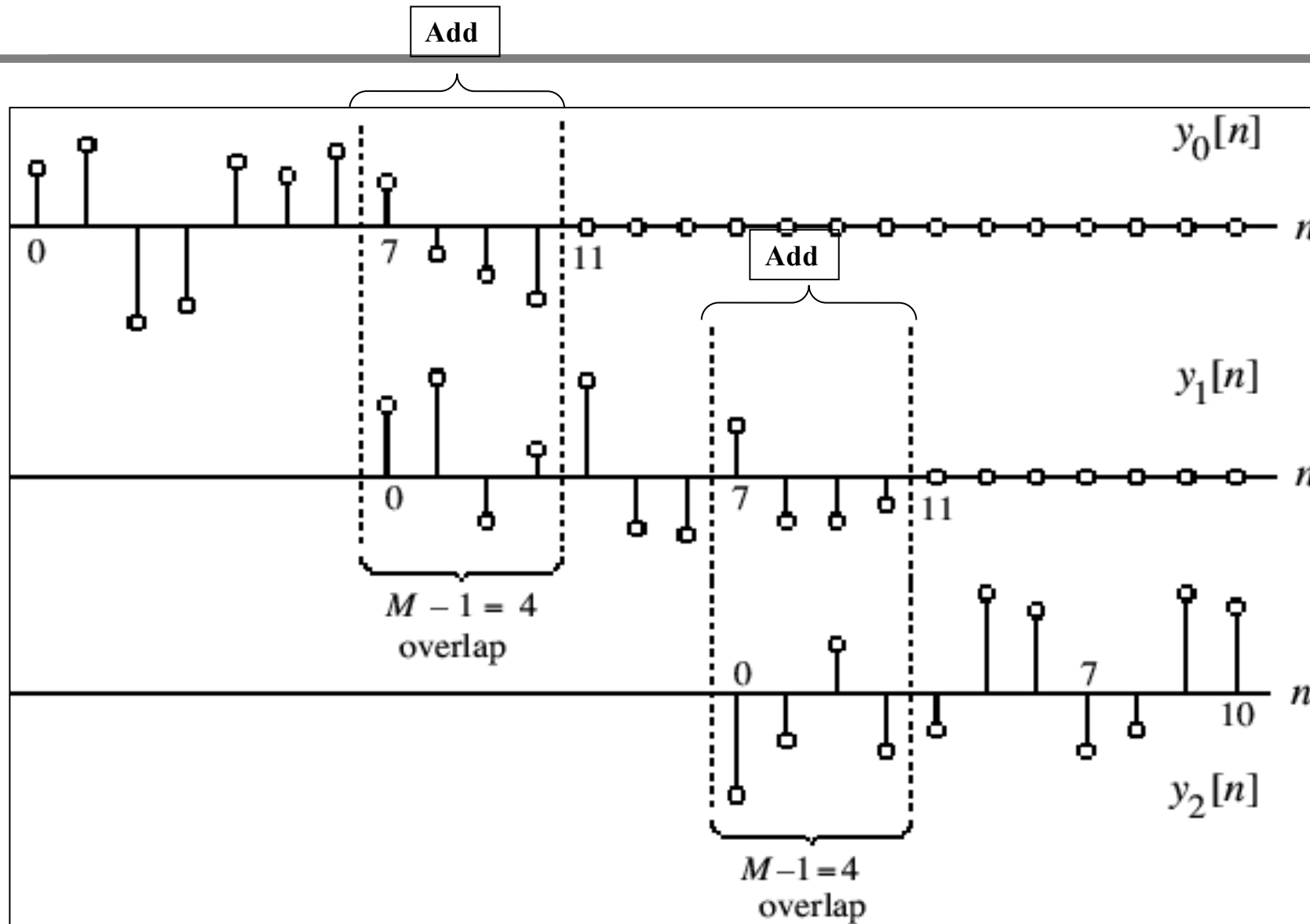
$$y_m[n] = x_m[n] \circledast h[n]$$

- These short convolutions can be implemented using the DFT-based approach.
- There is an **overlap of $M - 1$ samples** between these two short linear convolutions.

Segment $x[n]$ into subsequences



This figure illustrate Overlap-Add method for $M = 5$ and $N = 7$



$$\begin{aligned}
 y[n] &= y_0[n], & 0 \leq n \leq 6 \\
 y[n] &= y_0[n] + y_1[n-7], & 7 \leq n \leq 10 \\
 y[n] &= y_1[n-7], & 11 \leq n \leq 13 \\
 y[n] &= y_1[n-7] + y_2[n-14], & 14 \leq n \leq 17 \\
 y[n] &= y_2[n-14], & 18 \leq n \leq 20 \\
 &\vdots
 \end{aligned}$$

Overlap-Add Method



- The above procedure is called the overlap-add method since the results of the short linear convolutions overlap and the overlapped portions are added to get the correct final result.
- The function **fftfilt** can be used to implement the above method.
- Program 5_5 illustrates the use of **fftfilt** in the filtering of a noise-corrupted signal using a length-3 moving average filter.

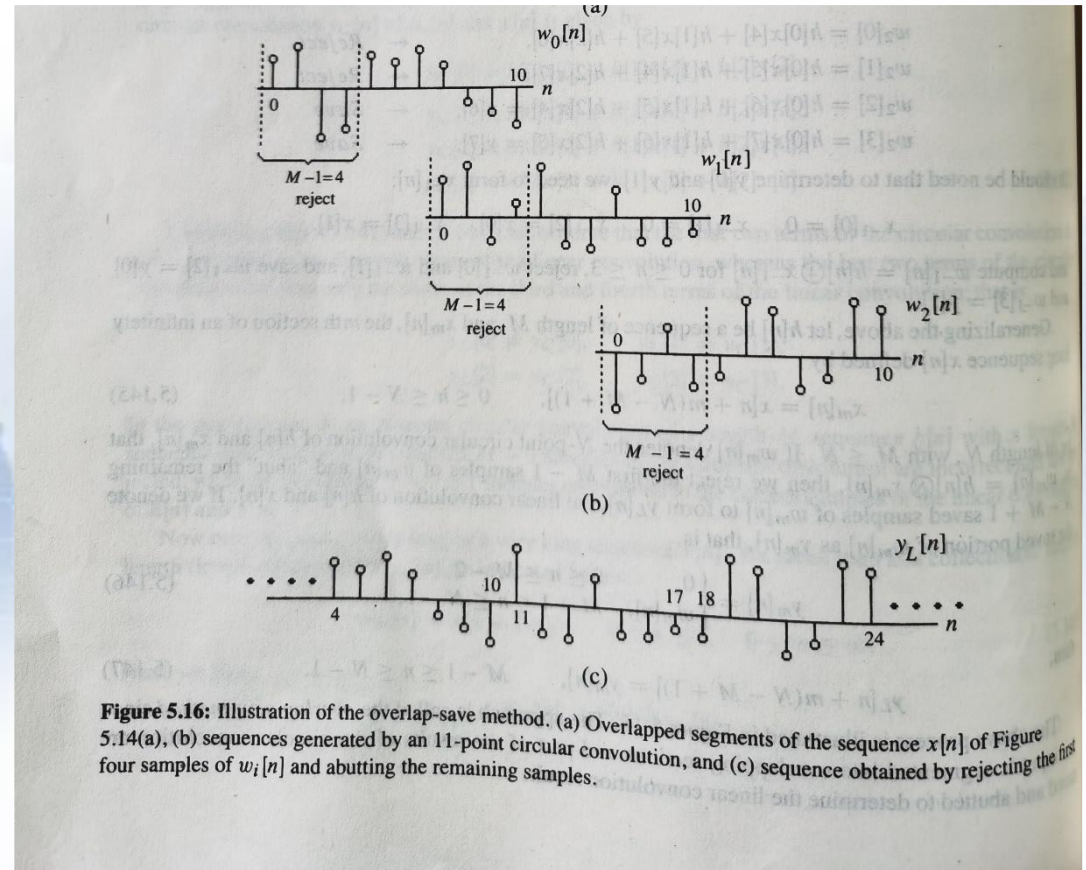
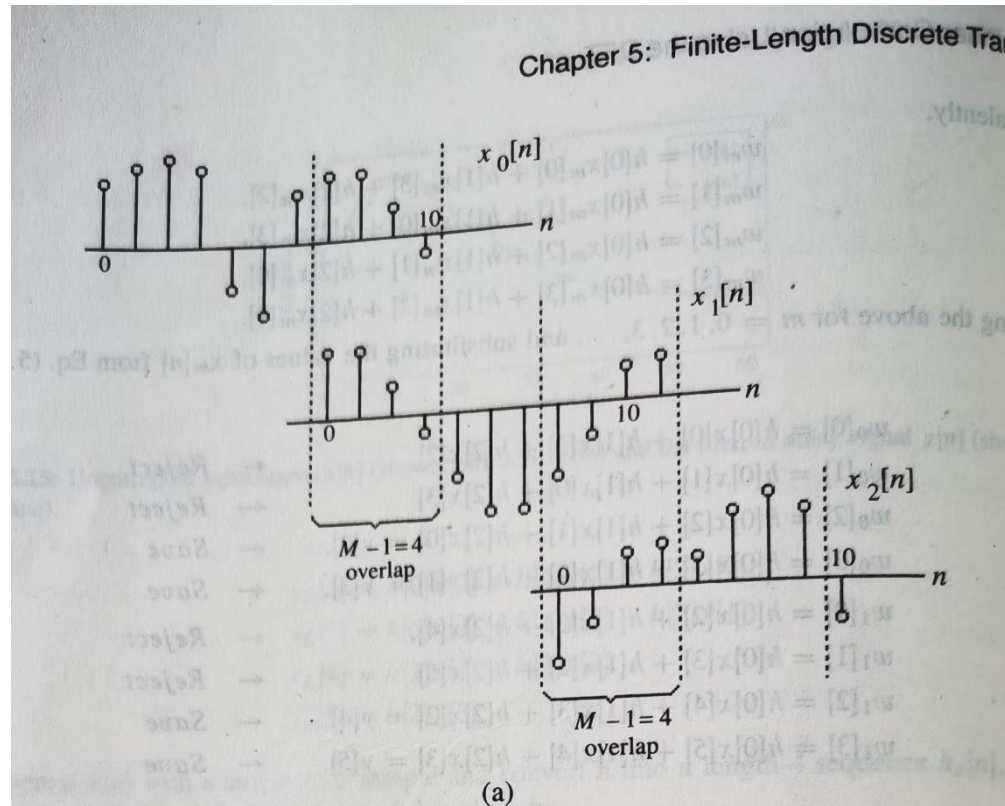
Overlap-Save Method



- At first, to segment $x[n]$ into **overlapping blocks** $x_m[n]$.
- Keeping the terms of the **circular convolution** of $h[n]$ with $x_m[n]$ that corresponds to the terms obtained by a linear convolution of $h[n]$ and $x_m[n]$.
- Throwing away the other part of the circular convolution.

Please look at P244, Fig5.16

Overlap-Save Method



Other Transforms



5.11 STFT

5.12 DCT

5.13 Haar Transform

Homework



- 5.4, 5.8 (a) (b), 5.13, 5.20, 5.25, 5.32, 5.36, 5.39, 5.40, 5.53, 5.64, 5.69, 5.76
- M5.2, M5.10, M5.11